MAST32001 Week 6 Summary

Zhi Zhou

2026-10-09

Lecture 11 started with Gibbs sampling. We draw a new value for each parameter while keeping the others at their latest values. We use the full conditional distribution, $\pi(\theta_j \mid \theta_{-j})$, for each update. The normal model with unknown μ and σ^2 was easier for me to follow. Normal and inverse-gamma priors give full conditionals that we can sample from directly. In R, we mainly need to keep track of the new values. Gibbs doesn't need a separate accept/reject step because each exact conditional update leaves the target distribution unchanged. We can also see it as a Metropolis-Hastings update with acceptance probability one. The choice of prior matters too. With $\mu \mid \sigma^2 \sim \mathcal{N}(m_0, \sigma^2/\kappa_0)$, the variance in the update for μ depends on the current value of σ^2.

The normal example with two variables showed that Gibbs can be slow even when its updates are correct. We compared $\rho = 0.2$ with $\rho = 0.99$. When the correlation is high, each conditional distribution is narrow. The chain takes small steps, and nearby draws stay similar for a long time. Both chains target the right distribution, but the second gives less new information from the same number of draws. Updating related parameters together or rewriting the model can help. HMC uses gradients and an added momentum variable to make longer moves through the posterior. Its total energy combines the negative log density with the kinetic energy of the momentum. This total stays the same under exact Hamiltonian dynamics. We use leapfrog steps to work out an approximate path, then a Metropolis step to correct for numerical error. The typical set was also useful to think about. For a normal distribution in many dimensions, most of the probability mass lies away from the point of highest density. Sampling needs to cover that mass.

NUTS chooses the path length by extending it until it starts to turn back. During warmup, Stan adjusts the step size and mass matrix. We write the data, parameters, and model, and Stan computes the gradients using automatic differentiation. We don't need to write the sampler ourselves. In the regression with the cars data, centring speed meant subtracting its average. The intercept then gave the expected stopping distance at the average speed. It was also less correlated with the posterior slope. I would still check the output before using the estimates. Effective sample size tells us how much information the draws contain. $\widehat{R}$ helps us check whether the chains agree. These checks don't prove that everything has worked. Divergent transitions mean the sampler has trouble following the path, often because of the shape of the posterior. A centered or non-centered version of the same hierarchical model can be easier or harder to sample from. Warmup is therefore a time to tune the sampler, not just a set of early draws to discard.

Lecture 12 covered variational inference (VI). We choose a simpler distribution q and adjust it

to approximate the posterior. We aim to make $\mathrm{KL}(q \,\|\, p(\theta \mid y))$ as small as possible. The key equation is $\log p(y) = \mathrm{ELBO} + \mathrm{KL}(q \,\|\, p(\theta \mid y))$. Since $\log p(y)$ doesn't change with q, raising the ELBO reduces the KL divergence. We don't need to calculate the marginal likelihood. In mean-field VI, the parts of q are independent. CAVI updates one part at a time while keeping the others fixed. For conjugate models, we can work out these updates directly. The regression example showed a problem with this approach. With correlated predictors, the means matched the exact posterior means, but the standard deviations were about three times too small in that example. Looking only at the means would miss this. I found the variance result harder to follow than the Gibbs updates. For a Gaussian posterior with precision matrix Λ, the exact variance of parameter j is $(\Lambda^{-1})_{jj}$. The best mean-field variance is $1/\Lambda_{jj}$. The first uses the inverse of the whole matrix; the second uses the reciprocal of one diagonal entry. They differ when the parameters are correlated. The direction of KL matters too. It penalizes q for putting too much probability where the posterior density is low. This helps explain why q is too narrow in this Gaussian example.

When we can't work out the VI updates directly, we can use Monte Carlo draws to estimate the ELBO and its gradients. The reparameterization trick starts with random noise that doesn't depend on the parameters of q. We then turn that noise into draws from q. This lets us calculate gradients through the steps that produce the draws. We also looked at how EM for a Gaussian mixture uses a lower bound. The E-step finds the conditional probabilities of the group labels that we don't observe. With these exact probabilities, the bound equals the log likelihood at the current model parameters. The M-step changes the parameters to raise the bound.

The mean-field variance result is the hardest part for me, while the Gibbs calculations were easier.